

Revised Generation Interconnection System Impact Study Report

For Queue Project Q20-1459

Monfort 100 kV
March 23, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1459. The Interconnection Customer proposes a 128.1 MW solar photovoltaic facility interconnecting at the Monfort 100 kV substation in the Indiana area. The request was received on April 17, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 389 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 19 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$9,000,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

COLUMBIA HILLS to PORTAGE UPPCO 100 KV: 243.8% loading (143 MVA) following the outage of the WAYNEDEALE SUBSTATION to MONFORT 230 KV; DFAX = 16.2%.

ILLINOIS ROAD to SOUTHPORT 100 KV Line: 195.2% loading (127 MVA) following the outage of the WAYNEDEALE to MONFORT 230 KV line; DFAX = 10.9%.

The SOUTHPORT to HART 100 KV is overloaded to 178.2% of its 152 MVA emergency rating for the loss of the WAYNEDEALE to MONFORT 230 KV. The project contributes 10.9% DFAX (14.0 MW) to this overload.

The PEAVINE to HOHENWALD 100 KV L is overloaded to 176.9% of its 147 MVA emergency rating for the loss of the WAYNEDEALE to STATE STREET SUBSTATION 230 KV. The project contributes 7.8% DFAX (10.0 MW) to this overload.

The WAYNEDEALE to INDEGO 100 KV is overloaded to 173.0% of its 242 MVA emergency rating for the loss of the WAYNEDEALE to MONFORT 230 KV. The project contributes 18.9% DFAX (24.2 MW) to this overload.

The HART to PORTAGE UPPCO 100 KV is overloaded to 166.1% of its 177 MVA emergency rating for the loss of the MONFORT to WAYNEDEALE 230 KV L. The project contributes 10.9% DFAX (14.0 MW) to this overload.

For the loss of WAYNEDEALE to MONFORT SUBSTATION 230 KV, the INDEGO to COLUMBIA HILLS 100 KV loads to 154.0% of its 294 MVA rating. Project DFAX: 18.7%.

ILLINOIS RD to HEBRON ROSEMEADE 100 KV: 151.4% loading (163 MVA) following the outage of the WAYNEDEALE to INDEGO 100 KV; DFAX = 14.4%.

TIEFORT to WINDER 100 KV Line 1: 146.6% loading (151 MVA) following the outage of the WAYNEDEALE to MONFORT SUBSTATION 230 KV; DFAX = 14.1%.

For the loss of WAYNEDEALE to MONFORT 230 KV, the TIEFORT to WINDER 100 KV Line 2 loads to 146.6% of its 151 MVA rating. Project DFAX: 14.1%.

For the loss of MONFORT to WAYNEDEALE 230 KV, the PORTAGE UPPCO to TIEFORT 100 KV Ckt 2 loads to 144.7% of its 151 MVA rating. Project DFAX: 14.1%.

The INDUSTRIAL PARK to SPY RUN 100 KV L is overloaded to 143.6% of its 161 MVA emergency rating for the loss of the WAYNEDEALE to SATE ST 230 KV. The project contributes 6.1% DFAX (7.9 MW) to this overload.

PORTAGE UPPCO to TIEFORT 100 KV L: 141.1% loading (154 MVA) following the outage of the MONFORT to WAYNEDEALE 230 KV L; DFAX = 14.1%.

MONFORT to WINDER 100 KV Line 2: 129.3% loading (182 MVA) following the outage of the MONFORT to WAYNEDEALE 230 KV; DFAX = 14.1%.

For the loss of WAYNEDEALE to INDEGO 100 KV, the MCKINLEY SUBSTATION to HEBRON ROSEMEADE SUBSTATION 100 KV line loads to 121.8% of its 134 MVA rating. Project DFAX: 10.7%.

The WINDER to MONFORT 100 KV #1 is overloaded to 120.1% of its 195 MVA emergency rating for the loss of the MONFORT to WAYNEDEALE 230 KV Line. The project contributes 14.1% DFAX (18.0 MW) to this overload.

For the loss of WAYNEDEALE SUBSTATION to STATE STREET 230 KV, the SEGS 5 to AEP PETTIT AVENUE 100 KV loads to 112.4% of its 265 MVA rating. Project DFAX: 7.7%.

For the loss of MONTPELIER ELECTRIC GENERATING STA to WABASH 500 KV, the MONFORT to WAYNEDEALE 230 KV line loads to 106.1% of its 1,323 MVA rating. Project DFAX: 37.3%.

The INDEGO to SATURN SUBSTATION 100 KV is overloaded to 102.6% of its 119 MVA emergency rating for the loss of the WAYNEDEALE to INDGO 100 KV. The project contributes 5.6% DFAX (7.2 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
COLUMBIA HILLS to PORTAGE UPPCO 100 KV	Rebuild line with higher-capacity conductor	\$3,295,000	9% (\$291,000)
ILLINOIS ROAD to SOUTHPORT 100 KV Line	Rebuild line with higher-capacity conductor	\$5,328,000	9% (\$480,000)
SOUTHPORT to HART 100 KV	Rebuild line with higher-capacity conductor	\$1,797,000	9% (\$162,000)
HOHENWALD to PEAVINE 100 KV L	Rebuild line with higher-capacity conductor	\$1,435,000	7% (\$97,000)
WAYNEDEALE to INDEGO 100 KV	Rebuild line with higher-capacity conductor	\$3,041,000	9% (\$272,000)
HART to PORTAGE UPPCO 100 KV	Rebuild line with higher-capacity conductor	\$4,161,000	9% (\$375,000)
COLUMBIA HILLS to IDEGO 100 KV	Rebuild line with higher-capacity conductor	\$2,254,000	9% (\$201,000)
ILLINOIS ROD to HEBRON ROSEMEADE 100 KV line	Rebuild line with higher-capacity conductor	\$1,985,000	9% (\$179,000)
TIEFORT to WINDER 100 KV Line 1	Rebuild line with higher-capacity conductor	\$5,316,000	14% (\$723,000)
TIEFORT to WINDER 100 KV Ckt 2	Rebuild line with higher-capacity conductor	\$6,476,000	14% (\$881,000)
PORTAGE UPPCO to TIEFORT 100 KV Ckt 2	Rebuild line with higher-capacity conductor	\$12,948,000	14% (\$1,762,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
INDUSTRIAL PARK to SPY RUN 100 KV L	Rebuild line with higher-capacity conductor	\$2,504,000	7% (\$180,000)
PORTAGE UPPCO to TIEFORT 100 KV L	Rebuild line with higher-capacity conductor	\$10,349,000	14% (\$1,408,000)
MONFORT to WINDER 100 KV Line	Rebuild line with higher-capacity conductor	\$2,250,000	14% (\$306,000)
MCKINLEY to HEBRON ROSEMEADE 100 KV	Rebuild line with higher-capacity conductor	\$1,936,000	12% (\$238,000)
WINDER to MONFORT 100 KV #1	Rebuild line with higher-capacity conductor	\$1,991,000	14% (\$271,000)
SEGS 5 to AEP PETTIT AVENUE 100 KV	Rebuild line with higher-capacity conductor	\$1,909,000	7% (\$139,000)
MONFORT to WAYNE DALE 230 KV line	Reconductor line	\$14,496,000	5% (\$684,000)
INDEGO to SATURN SUBSTATION 100 KV	Reconductor line	\$2,759,000	13% (\$351,000)
	Total allocated to project		\$9,000,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.